

Question			Answer/Indicative content	Marks	Part marks and guidance	
1		i	$X \sim B(10, 0.35)$	M1	or $0.35^5 \times 0.65^5$	
		i	$P(5 \text{ accessing internet}) = \binom{10}{5} \times 0.35^5 \times 0.65^5$	M1	For $\binom{10}{5} \times p^5 \times q^5$	With $p + q = 1$ Also for 252×0.0006094
		i	$= 0.1536$	A1	cao	Allow 0.15 or better NB 0.153 gets A0 See tables at the website http://www.mei.org.uk/files/pdf/formula_book_mf2.pdf
		i	OR			
		i	from tables = $0.9051 - 0.7515 = 0.1536$	M2	For $0.9051 - 0.7515$	
		i		A1	cao	Accept 0.25 or better – allow 0.248 or 0.249 Calculation of individual probabilities gets B2 if fully correct 0.25 or better, otherwise B0.
		i	$P(X \geq 5) = 1 - P(X \leq 4)$			
		i	$= 1 - 0.7515$	M1	For 0.7515	
		i	$= 0.2485$	A1	cao	
		i	$E(X) = np = 10 \times 0.35$	M1	For 10×0.35	If any indication of rounding to 3 or 4 allow M1A0

Examiner's Comments

Again many fully correct responses were seen. Candidates usually used the correct table but a common wrong answer was $1 - P(X \leq 5)$ rather than $1 - P(X \leq 4)$. Some candidates used the lengthy method of finding the individual probabilities of 5 or more and then adding, sometimes successfully but in many cases with errors.

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		i	= 3.5	A1	cao Examiner's Comments Almost all candidates multiplied 0.35 by 10, but about 20% of them either rounded to 4 or truncated to 3, thus losing the second mark.	
		ii	Let $X \sim B(20, 0.35)$	9		
		ii	Let p = probability of a customer using the internet (for population)	B1	For definition of p in context	Minimum needed for B1 is p = probability of using internet. Allow $p = P(\text{using internet})$ Definition of p must include word probability (or chance or proportion or percentage or likelihood but NOT possibility). Preferably as a separate comment. However can be at end of H_0 as long as it is a clear definition ' p = the probability of using internet', Do NOT allow ' p = the probability of using internet is different'

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		ii	$H_0: p = 0.35$	B1	For H_0	Allow $p = 35\%$, allow only p or θ or π or ρ. However allow any single symbol if defined (including x) Allow $H_0 = p = 0.35$, Allow H_0: $p = 7/20$ or $p = 35/100$ Allow NH and AH in place of H_0 and H_1 Do not allow $H_0 : P(X = x) = 0.35$ Do not allow $H_0: = 0.35, = 35\%$, $P(0.35), p(x) = 0.35, x = 0.35$ (unless x correctly defined as a probability) Do not allow H_0 and H_1 reversed For hypotheses given in words allow Maximum B0B1B1 Hypotheses in words must include probability (or chance or proportion or percentage) and the figure 0.35 oe Thus eg $H_0 : p(\text{using internet}) = 0.35, H_1 : p(\text{using internet}) \neq 0.35$ gets B0B1B1
		ii	$H_1: p \neq 0.35$	B1	For H_1	Allow '$p < 0.35$ or $p > 0.35$' in place of $p \neq 0.35$
		ii	H_1 has this form because the test is to investigate whether the proportion is different, (rather than lower or higher).	E1		Do not allow if H_1 wrong.

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		ii	$P(X \geq 10)$	B1	For notation $P(X \geq 10)$ or $P(X > 9)$ or $1 - P(X \leq 9)$ (as long as no incorrect notation)	This mark may be implied by 0.1218 as long as no incorrect notation. No further marks if point probs used - $P(X = 10) = 0.0686$ (do not even give the notation mark for correct notation) DO NOT FT wrong H_1 , but see extra notes
		ii	$= 1 - 0.8782 = 0.1218$	B1*	For 0.1218 Allow 0.12	Or for $1 - 0.8782$ Indep of previous mark
		ii	$> 2.5\%$	M1* dep	For comparison with 2.5%	
		ii	So not significant.	A1*		Allow 'accept H_0 ' or 'reject H_1 '
		ii	Conclude that there is not enough evidence to indicate that the probability is different. (Must state 'probability', not just 'p')	E1* dep on A1		Must include 'sufficient evidence' or something similar such as 'to suggest that' ie an element of doubt either in the A or E mark.
		ii	ALTERNATIVE METHOD FOR FINAL 5 MARKS			
		ii	Critical region method			Do not insist on correct notation as candidates have to work out two probabilities for full marks. If only upper tail of CR given (or only upper tail justified), allow max 4/5 for final 5 marks.
		ii	LOWER TAIL $P(X \leq 2) = 0.0121 < 2.5\%$ $P(X \leq 3) = 0.0444 > 2.5\%$	B1	For either probability	

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		ii	UPPER TAIL $P(X \geq 11) = 1 - P(X \leq 10) = 1 - 0.9468 = 0.0532 > 2.5\%$ $P(X \geq 12) = 1 - P(X \leq 11) = 1 - 0.9804 = 0.0196 < 2.5\%$	B1	For either probability	
		ii	So critical region is {0, 1, 2, 12, 13, 14, 15, 16, 17, 18, 19, 20}	M1*	For either probability can depend on at least one correct comparison with 2.5% Examiner's Comments Most candidates were able to identify that this was a two-tailed test and were able to correctly state the null and alternative hypotheses. However, some candidates failed to define p and others failed to explain why it was two-tailed. Some of the weaker candidates used poor notation when defining their hypotheses. Rather more candidates used the critical region method than finding $P(X \geq 10)$. However, those who used the probability method were generally more successful. Those who tried to find the critical region often included either 3 or 11 and so lost the final three marks. Unfortunately, a significant number of candidates made comparisons with 5% instead of 2.5%, or omitted the comparison altogether and so again lost the last three marks. A disappointing number found $P(X=10)$ thus losing all of the final 5 marks. It was pleasing to see that the majority of candidates did however realise that justification, with	No marks if CR not justified Condone {0, 1, 2, 12, ... 20}, $X \leq 2$, $X \geq 12$, or but not $P(X \leq 2)$ etc

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					probabilities, is needed whichever method they employ. Conclusions, for those who get this far, were usually correct. However care should be taken to explain <i>in words</i> their findings including an <i>element of doubt</i> in their conclusion. Those answering by the critical region method should be aware that '10 is not in CR' is not enough, they also need to add 'insufficient evidence to reject the null hypothesis' and then go on to give an answer in context.	
		ii	So not significant	A1*		
		ii	Conclude that there is not enough evidence to indicate that the probability is different.	E1* dep on A1		NB If CR found correctly then P(X = 10) subsequently found but cand says '10 not in CR' then allow up to all last five marks. If do not say '10 not in CR' allow none of last five marks
		iii	$0.0022 < 2.5\%$	2		
		iii	So reject H_0 , Significant.	B1	For either reject H_0 or significant, dep on correct comparison	

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		iii	Conclude that there is enough evidence to indicate that the probability is different.	E1* dep	Dep on good attempt at correct hypotheses in part (ii) Examiner's Comments Under half of the candidature scored either mark in this question. Many did not attempt it. A disappointing proportion compared with 5% even though they had correctly compared with 2.5% in part (ii). A further significant proportion failed to correctly state their conclusion within the context of the question.	If they have $H_1: p > 0.35$, allow SC1 if all correct including comparison with 5%.
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
2		i	Because if people cannot make a correct identification, then the probability that they guess correctly will be 0.5	E1	For idea of a guess or 'chosen at random'	NB The question includes the sentence 'She suspects that people do no better than they would by guessing.', so this on its own does not get the mark for the idea of a guess
		i	For 'equally likely to guess right or wrong' or 'two outcomes with equal probability' or '50:50 chance of success' or 'right one in two occasions on average' or 'two (equally likely) outcomes' etc	E1	For idea of two outcomes <u>Examiner's Comments</u> The wording of the researcher's theory appeared to cause confusion for some of the candidates throughout the question. This was translated into some poorly worded explanations and conclusions in all three parts of the question. Good comprehension skills are required in this type of question and, unfortunately, these skills were not always in evidence. Many candidates scored both marks. Unfortunately a good proportion lost either the first or the second mark by not mentioning 'guess' or only including it when they quoted the question or not mentioning, in any form, the idea of the two possible outcomes. Some candidates simply just re-stated the null hypothesis in words.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	'Because people may do better than they would by guessing' or similar	B1	For idea of selecting correctly / identifying / knowing Examiner's Comments The wording of the researcher's theory appeared to cause confusion for some of the candidates throughout the question. This was translated into some poorly worded explanations and conclusions in all three parts of the question. Good comprehension skills are required in this type of question and, unfortunately, these skills were not always in evidence. This was not as well answered as part (i). There was a failure to distinguish between guessing and being able to identify between the two types of water. A lot of candidates lost the mark because they gave the reason for the alternative hypothesis as '13 people out of 20 in the researcher's sample identified correctly' which of course is not a valid reason.	No marks if answer implies that it is because there are over half in the sample who make a correct identification
		iii	$P(X \geq 13) = 1 - P(X \leq 12) = 1 - 0.8684 = 0.1316$	M1	For notation $P(X \geq 13)$ or $P(X > 12)$ or $1 - P(X \leq 12)$	Notation $P(X = 13)$ scores M0. If they have the correct $P(X \geq 13)$ then give M1 and ignore any further incorrect notation.
		iii		B1*	For 0.1316	Or for $1 - 0.8684$ indep of previous mark
		iii	$0.1316 > 0.05$	M1* dep	For comparison with 5%	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	So not significant	A1*		Allow 'accept H_0 ' or 'reject H_1 '
		iii	There is insufficient evidence to suggest that people can make a correct identification.	E1* dep	NB Point probabilities score zero.	Must include 'insufficient evidence' or something similar such as 'to suggest that' ie an element of doubt either in the A or E mark. Must be in context to gain E1 mark. Do not allow 'sufficient evidence to suggest proportion making correct identification is 0.5' or similar
		iii	ALTERNATIVE METHOD – follow method above unless some mention of CR seen		Must see some reference to CR to gain any marks	
		iii	Critical region method UPPER TAIL $P(X \geq 14) = 1 - P(X \leq 13) = 1 - 0.9423 = 0.0577 < 5\%$	B1	For either probability	Do not insist on correct notation as candidates have to work out two probabilities for full marks.
		iii	$P(X \geq 15) = 1 - P(X \leq 14) = 1 - 0.9793 = 0.0207 < 5\%$	M1*	For a correct comparison with 5%	Allow comparison in form of statement 'critical region at 5% level is ...'
		iii	So critical region is {15,16,17,18,19,20}	M1* dep	cao dep on the two correct probabilities	No marks if CR not justified Condone {15, ... 20}, $X \geq 15$, oe but not $P(X \geq 15,)$ etc
		iii	13 not in CR so not significant	A1*	Must include '13 not in CR'	Allow 'accept H_0 ' or 'reject H_1 '

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	There is insufficient evidence to indicate that people can make a correct identification.	E1* dep on A1	Ignore any work on lower critical region <u>Examiner's Comments</u> The wording of the researcher's theory appeared to cause confusion for some of the candidates throughout the question. This was translated into some poorly worded explanations and conclusions in all three parts of the question. Good comprehension skills are required in this type of question and, unfortunately, these skills were not always in evidence. The most successful way of approaching this hypothesis test was to compare $P(X \geq 13)$ with the significance level. Several of the candidates, who used this method failed to gain the final mark due to not putting the explanation in the context of the question. Other candidates used incorrect probabilities, usually $P(X \geq 12)$ or $P(X \geq 14)$. Candidates who used the critical region method normally gained the first two marks but then many of them failed to gain any more marks – usually because they had included 14 in the critical region. Unfortunately some candidates started looking at the two probabilities necessary for the critical region but made no mention of the critical region, or	NB If CR found correctly, then $P(X=13)$ subsequently found, but cand says '13 not in CR' then allow up to all five marks. If do not say '13 not in CR' allow no marks NOTE RE OVER-SPECIFICATION OF ANSWERS If answers are grossly over-specified, deduct the final answer mark in every case. Probabilities should also be rounded to a sensible degree of accuracy. In general final non probability answers should not be given to more than 4 significant figures. Allow probabilities given to 5 sig fig. PLEASE HIGHLIGHT ANY OVER-SPECIFICATION Please note that there are no G or E marks in scoris, so use B instead Additional notes Comparison with 95% method If 95% seen anywhere then M1 for $P(X \leq 12)$ B1 for 0.8684 M1* for comparison with 95% dep on second B1 A1* for not significant oe E1* Comparison with 95% CR method If 95% seen anywhere then B1 for 0.9423 or 0.9793 M1 for correct comparison

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					critical value, so did not gain any marks. It is pleasing to report, on the other hand, that very few candidates tried to use point probabilities. However, although full marks could be obtained by comparing 0.8684 with 95%, many candidates either compared with 5% or made no explicit comparison at all - such candidates were unable to gain any credit.	with 95% M1dep for correct CR provided both probs correct then follow mark scheme for CR method Smallest critical region method: Smallest critical region that 13 could fall into is {13, 14, 15, 16, 17, 18, 19, 20} gets B1 and has size 0.1316 gets B1, This is > 5% gets M1*, A1*, E1* as per scheme NB These marks only awarded if 13 used, not other values. Use of k method with no probabilities quoted: This gets zero marks. Use of k method with one probability quoted: Mark as per scheme Line diagram method and Bar chart method No marks unless correct probabilities shown on diagram, then mark as per scheme..
			Total	8		

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3		i	(A) $X \sim B(15, 0.85)$	M1	For $0.85^{12} \times 0.15^3$	With $p + q = 1$
		i	$P(\text{exactly 12 germinate}) = \binom{15}{12} \times 0.85^{12} \times 0.15^3$	M1	For $\binom{15}{12} \times p^{12} \times q^3$	Also for 455×0.00048
		i	$= 0.2184$	A1	CAO	Allow 0.22 or better
		i	OR	OR		See tables at the website
		i	from tables: $0.3958 - 0.1773$	M2	For $0.3958 - 0.1773$	http://www.mei.org.uk/files/pdf/formula_book_mf2.pdf
		i	$= 0.2185$	A1	CAO	
					Examiner's Comments This was generally very well answered.	
		ii	(B) $P(X < 12) = P(X \leq 11) = 0.1773$	M1	For $P(X \leq 11)$ or $P(\leq 11)$ (With no extras)	Accept 0.18 or better Calculation of individual probabilities gets B2 if fully correct in range 0.177 to 0.18, otherwise B0.
		ii		A1	CAO (as final answer) May see alternative method: $0.3958 - 0.2185 = 0.1773$ $0.3958 -$ their wrong answer to part (i) scores M1A0 Examiner's Comments Although most candidates answered this correctly, some gave $P(X \leq 12)$ rather than $P(X \leq 11)$, and some found the required probability but then subtracted it from 1.	
		iii	Let p = probability of a seed germinating (for the population)	B1	For definition of p in context	
		iii				See below for additional notes
		iii	$H_0: p = 0.85$	B1	For H_0	

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		iii	$H_1: p < 0.85$	B1	For H_1	
		iii	H_1 has this form because the test is to investigate whether the proportion of seeds which germinate is lower.	E1	Dep on < 0.85 used in H_1 Do not allow just 'Germination rate will be lower' or similar. Examiner's Comments Most candidates wrote down the correct hypotheses using the correct notation. It is encouraging to report that rather more candidates gave a correct definition of p than was the case in previous years.	For use of 0.15 as $P(\text{not germinating})$, contact team leader E0 for simply stating H_1 in words
		iv	Let $X \sim B(20, 0.85)$			
		iv	$P(X \leq 13) = 0.0219$	M1*	For probability (provided not as part of finding	No further marks if point probs used - $P(X = 13) = 0.0160$
		iv			$P(X = 13)$) Ignore notation	DO NOT FT wrong H_1 , but see extra notes
		iv	$0.0219 > 1\%$	M1*dep	For comparison	Allow 'accept H_0 ' or 'reject H_1 '
		iv	So not enough evidence to reject H_0 .	A1*	For not significant oe	Must include 'sufficient evidence' or something similar such as 'to suggest that' ie an element of doubt either in the A or E mark.
		iv	Not significant.			
		iv	Conclude that there is not enough evidence to indicate that the proportion of seeds which have germinated has decreased.	E1*dep	For conclusion in context Must mention decrease, not just change	
		iv	ALTERNATIVE METHOD – follow method above unless some mention of CR seen Critical region method LOWER TAIL			No marks if CR not justified Condone $\{0, 1, 2, \dots, 12\}$, $X \leq 12$, oe but not $P(X \leq 12)$ etc

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		iv	$P(X \leq 13) = 0.0219 > 1\%$	M1	For either probability	
		iv	$P(X \leq 12) = 0.0059 < 1\%$			
		iv	So critical region is $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$	A1	cao dep on at least one correct comparison with 1%	Could get M1A0A1E1 if poor notation for CR
		iv	13 not in CR so not significant	A1*		Do not allow just '13 not in CR'
		iv	There is insufficient evidence to indicate that the proportion of seeds which have germinated has decreased.	E1*dep	Examiner's Comments Those candidates who calculated $P(X \leq 13)$ were generally more successful than those using a critical region method. Those who used the latter method often got the critical region wrong, thereby losing credit. In general conclusions were given more clearly than in previous sessions, although not always in context. There was also rather less use of point probabilities than in the past.	- Must say 'not significant' or accept H_0 or similar
		v	$33 < 35$	M1	For comparison	Allow '33 lies in the CR'
		v	So there is sufficient evidence to reject H_0	A1*		Must include 'sufficient evidence' or something similar such as 'to suggest that' ie an element of doubt either in the A or E mark.
		v				Do not FT wrong H_1 : In part (iv) ignore any interchanged H_0 and H_1 seen in part (ii)
		v	Conclude that there is enough evidence to indicate that the proportion of seeds which have germinated has decreased.	E1*dep	For conclusion in context	

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		v			Must mention decrease, not just change Examiner's Comments Many candidates, despite having answered the previous part correctly, reverted to point probabilities in this part, using their calculator to find $P(X = 33)$. This of course gained no credit. Others made a correct comparison ($33 < 35$) but were not always sure what this meant in the context of the test.	If use a calculator to find $P(X \leq 33) = 0.000661$ and compare with 1% then B2 for $P(X \leq 33) = 0.000661 < 0.01$ so reject H_0 then final E1 as per scheme.
		vi	For $n = 3$, $P(X \leq 0) = 0.0034 < 0.01$	M1	For $P(X \leq 0) = 0.0034$	Allow 0.003
		vi	For $n = 2$, $P(X \leq 0) = 0.0225 > 0.01$	M1	For $P(X \leq 0) = 0.0225$	
		vi	So the least value of n for which the critical region is not empty and thus H_0 could be rejected is 3.	A1	CAO	Condone just ' $n = 3$ ' for final A mark dep on both M marks
		vi				If wrong H_1 allow max M2A0 if correct probabilities seen.
		vi	ALTERNATIVE METHOD using logs			
		vi	$0.15^n < 0.01$	M1		
		vi	$n > \log 0.01 / \log 0.15$	M1		
		vi	$n > 2.427$			

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		vi	Least $n = 3$	A1	Examiner's Comments Most candidates who knew how to tackle this question wrote down 'for $n = 3$, $P(X = 0) = 0.0034 < 0.01$ '. However many did not then justify their answer by writing down $P(X = 0)$ for $n = 2$ and thus only gained one mark. There were very few successful attempts using logarithms.	
			Total	19		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		i	(A) $X \sim B(20, 0.78)$ $P(\text{Exactly 19 cured}) = \binom{20}{19} \times 0.78^{19} \times 0.22^1$	M1	For $0.78^{19} \times 0.22^1$	Allow M2A0 for linear interpolation from tables leading to $0.9918 - 0.9488 = 0.0430$ But zero for use of tables with 0.8 leading to $0.9885 - 0.9308 = 0.0577$
		i		M1	For $\binom{20}{19} \times p^{19} \times q^1$	With $p + q = 1$ Also for 20×0.00196
		i	$= 0.0392$ (0.039197)	A1	CAO	Allow 0.039 or better Condone 0.03919 but not 0.0391
					Examiner's Comments This was generally very well-answered.	
		i	(B) $P(\text{Exactly 20 cured}) = \binom{20}{20} \times 0.78^{20} \times 0.22^0 = 0.0069$	M1	For 0.78^{20} oe	Allow M2 for 0.9488 for linear interpolation from tables or M1 for $1 - 0.9918 = 0.0082$ and second M1 for correct FT using answer to (i)(A) Zero for use of $p = 0.8$ here
		i	$P(\text{At most 18 cured}) = 1 - (0.0069 + 0.0392)$	M1	For $P(19) + P(20)$	Not necessarily correct, but both attempts at binomial, including coefficient in (i) and no extra terms (such as $P(X = 18)$) Condone use of $p = 0.8$
		i	$= 0.954$ (0.95385)	A1	CAO	Allow 0.95 with working
					Examiner's Comments Although around two-thirds of candidates answered this correctly, some candidates included $P(X=18)$ in their method and thus were only able to gain 1 mark out of 3.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	(C) $E(X) = np = 20 \times 0.78 = 15.6$	B1	CAO Examiner's Comments The majority of the candidates found this part straightforward, but a minority lost the mark when they rounded their final answer to 15 or 16.	Do not allow final answer of 15 or 16 even if correct 15.6 given earlier
		ii	Let $X \sim B(20, 0.78)$	B1	For definition of p	In context See below for additional notes
		ii	$H_0: p = 0.78$	B1	For H_0	
		ii	$H_1: p > 0.78$	B1	For H_1	No further marks if point probabilities used
		ii	$P(X \geq 19) = 0.0392 + 0.0069$	B1	For NOTATION $P(X \geq 19)$ or $P(X > 18)$ or $1 - P(X \leq 18)$ or $1 - P(X < 19)$	Notation $P(X = 19)$ scores B0. If they have the correct $P(X \geq 19)$ then give B1 and ignore any further incorrect notation.
		ii	$= 0.0461$	B1*	CAO For 0.0461 allow 0.0462	FT answer to (i)B for following three marks provided based on $1 - (P(19) + P(20))$
		ii	$0.0461 > 1\%$	M1* dep	For comparison with 1%	Dep on sensible attempt at $P(X \geq 19)$
		ii	So not significant.	A1		Allow 'accept H_0 ' or 'reject H_1 '

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		ii	Conclude that there is not enough evidence to suggest that the new drug is more effective than the old one.	E1		Must include 'insufficient evidence' or something similar such as 'to suggest that' ie an element of doubt either in the A or E mark. Must be in context to gain E1 mark. Do NOT allow 'sufficient evidence to suggest proportion cured is 0.78' or similar 99% method: $P(X \leq 18) = 0.9539$ B1B1* CAO $0.9539 < 99\%$ M1* then as per scheme
		ii	ALTERNATIVE METHOD FOR FINAL 5 MARKS		If combination of methods used, mark both and give higher mark.	No further marks if point probabilities used
		ii	$P(X \geq 19) = 0.0461 > 1\%$	3	For either probability	Do not insist on correct notation as candidates have to work out two probabilities for full marks.
		ii	$P(X \geq 20) = 0.0069 < 1\%$	M1	For at least one comparison with 1%	Allow comparison in form of statement 'critical region at 1% level is ...'
		ii	So critical region is {20}	B1*	CAO dep on the two correct probabilities	No marks if CR not justified Condone $X \geq 20$, $X = 20$, oe but not $P(X \geq 20)$, etc
		ii	(19 not in CR so) not significant.	A1* dep	Dep on correct CR	Allow 'accept H_0 ' or 'reject H_1 '

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		ii	Conclude that there is not enough evidence to suggest that the new drug is more effective than the old one.	E1* dep	Ignore any work on lower critical region Examiner's Comments In recent years, candidates have been doing better on hypothesis test questions than in the past, and this was again the case this year. Many fully correct responses were seen. Most candidates scored the first three marks for the hypotheses, with most now knowing that they need to define p. The vast majority of successful candidates used the probability method, finding $P(X \geq 19)$ and then comparing this to 1%. It was pleasing to see that most candidates gave their final answer in context and with an element of doubt stating something to the effect of 'there is not enough evidence to suggest that...'. Those who tried to use the critical region method were less successful on the whole. Again some tried to use point probabilities, being able to gain only the first three marks for the hypotheses. A few candidates tried to use tables and there full marks available for correct interpolation from tables.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	With a 5% significance level rather than a 1% level, the null hypothesis would have been rejected. OR: 'there would be enough evidence to suggest that the new drug is more effective than the old one.'	B1*	oe	FT their probability from (ii) but NO marks if point probabilities used There must be a sensible attempt to use $P(X = 19) + P(X = 20)$ or must have correct CR.
		iii	This is because $0.0461 < 5\%$	B1* dep	oe Examiner's Comments Candidates who gained more or less full marks in part (ii) tended to gain full marks in this part. In this part no marks were available if point probabilities were used.	Dep on correct answer of 0.0461 compared with 5% or 0.9539 compared with 95% or correct CR.
			Total	17		

Question			Answer/Indicative content	Marks	Part marks and guidance	
5		i	(A) $X \sim B(16, 0.1)$	Enter text here.		
		i	$P(X=3) = 0.25^3 \times 0.75^{16-3}$ 0.1423	M1	For $0.1^3 \times 0.9^{13}$	With $p + q = 1$
		i		M1	For $\binom{16}{3} \times p^3 \times q^{13}$	Also for $560 \times 0.000254..$ Allow 0.14 or better
		i		A1	CAO	
		i	Or: From tables	M2	For $0.9316 - 0.7892$	
		i	$P(X \leq 3) - P(X \leq 2) =$ $0.9316 - 0.7892 = 0.1424$	A1	CAO	
		i			Examiner's Comments This was very well answered.	
		i	$(B) P(X \geq 3) = 1 - P(X \leq 2)$ $= 1 - 0.7892 = 0.2108$	M1	For 0.7892	If calculating $P(X = 0) + P(X = 1) + P(X = 2)$ allow M1 for 0.79 or better and A1 for 0.21 or better.
		i		A1	CAO	
					Examiner's Comments Again this was well answered, usually by use of tables, although some candidates did calculate the three probabilities, add and subtract from 1. A few candidates forgot to subtract from 1, and a few just subtracted $P(X = 2)$ from 1.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	(C) Expected number = $16 \times 0.1 = 1.6$	B1	Examiner's Comments The majority of the candidates found this part straightforward, but a small minority lost the mark when they rounded their final answer to 1 or 2.	Do not allow final answer of 1 or 2 even if correct 1.6 given earlier
		ii	Let p = probability of a randomly chosen person using 1234 as their PIN (in the population) $H_0: p = 0.1$	B1	For definition of p (in context)	Do NOT allow number in place of probability. See below for additional notes
		ii	$H_1: p < 0.1$	B1		
		ii		B1	For H_0 For H_1	
		ii	The alternative hypothesis has this form as the advertising campaign aims to reduce the proportion of the population who use 1234 as their PIN.	B1	Dep on < 0.1 used in H_1 Do Not allow just 'proportion will be lower' or similar. Minimum needed for B1 is p = probability of using 1234. Allow $p = P(\text{using 1234})$ Definition of p must include word probability (or chance or proportion or percentage or likelihood but NOT possibility, number or amount). Preferably given as a separate comment. However can be at end of H_0 as long as it is a clear definition 'p = the probability of using 1234.' Do NOT allow 'p = the probability of using 1234 is different' Allow $p=10\%$, allow only p or θ or π or ρ . However allow any single symbol if defined (including x) Allow $H_0 = p=0.1$, Allow $H_0 : p=1/10$ Allow NH and AH in place	For use of 0.9 as P(do not use 1234), contact team leader. E0 for simply stating H_1 in words Condone number instead of proportion.

Question			Answer/Indicative content	Marks	Part marks and guidance	
					of H_0 and H_1 Do not allow $H_0 : P(X = x) = 0.1$ Do not allow $H_0 : = 0.1$, $= 10\%$, $P(0.1)$, $p(x) = 0.1$, $x = 0.1$ (unless x correctly defined as a probability) Do not allow H_0 and H_1 reversed For hypotheses given in words allow Maximum B0B1B1 Hypotheses in words must include probability (or chance or proportion or percentage) and the figure 0.1 oe Thus eg $H_0 : P(\text{using 1234}) = 0.1$, $H_1 : P(\text{using 1234}) < 0.1$ gets B0B1B1 Examiner's Comments As in recent years, candidates did well on this part, with over 80% gaining at least 3 marks out of 4. Most candidates scored the first two marks for the hypotheses, with many knowing that they needed to define p, thus scoring the third mark. A valid explanation of the reason for the form of the alternative hypothesis was usually given, even if not always very well worded.	
		iii	(A) For $n = 20$, $P(X \leq 0) = 0.1216$	M1*	For sight of 0.1216	Condone $P(X = 0)$ in place of $P(X \leq 0)$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	0.1216 > 0.10	*M1dep	For > 0.10 or > 10% Do NOT FT wrong H_1	Need to see a comparison with 0.1 or 10% explicitly, not just mentioning significance level. Allow SC2 for clearly indicating use of $B(20, 0.1)$ but with no mention of 0.1216 with convincing reasoning and final answer correct Allow CR is empty but NOT CR is zero
		iii	So no point in carrying out the test as H_0 could not be rejected (even if nobody in the sample uses 1234 as their PIN). oe	A1	or state 'There is no critical region' oe For A1 need $P(X \leq 0)$ or $P(X = 0)$ somewhere oe Examiner's Comments Only about half of the candidates scored any marks here at all. Many candidates did not use any numbers so could not gain all the marks, but were awarded special case 2 if they gave a very convincing explanation. Some of those that did state that $P(X \leq 0) = 0.1216$ then failed to show a comparison with 10% or 0.1 and so only scored 1 mark.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	(B) Lowest value of k is 13	B1	Or 13% Examiner's Comments Those candidates who had 0.1216 in the previous part usually gave the correct answer of 13%. Some who did not get marks in the previous part did give the correct answer, so they probably simply did not know how to verbalise the previous answer. However under half scored the mark, some by simply failing to round to an integer.	
		iv	$P(X \leq 2) = 0.0530$	B1	For use of $P(X \leq 2)$ only	No marks unless H_1 correct
		iv	$0.0530 > 0.05$	M1	For comparison of 0.0530 with 5%	If B0 then no further marks
		iv	So not significant. Do not reject H_0	A1*	Also allow $P(X \leq 2) >$	Allow 'accept H_0 ' or 'reject H_1 '

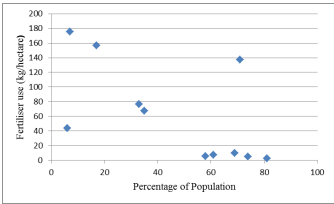
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	Conclude that there is not enough evidence to support the suggestion that the advertising campaign has been successful. Reminder: When you mark this question part, if you 'fit to height' you can check the last page for working or mark it BP if there none	*EI dep	0.05, $(P(X \leq 1) < 0.05)$ so CR is $\{0, 1\}$ for first two marks then A1E1 as usual Condone 'number of people' in conclusion Examiner's Comments There were many good, clear answers to this part of the question but there were still a good proportion of candidates that were tempted to use point probabilities. A significant number who did use the correct probability (or probabilities if using a critical region method) failed to give the conclusion of the test in context. Some lost the final mark for commenting that the proportion had not changed instead of had not reduced and some gave a conclusion which was too assertive.	Must include 'insufficient evidence to suggest that' or something similar i.e. an element of doubt either in the A or E mark.
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6		i	$P(Y = 76) = P\left(\frac{75.5 - 76}{12} \leq Z \leq \frac{76.5 - 76}{12}\right)$ $= P(-0.04166... < Z < 0.04166...)$ $= \Phi(0.04166...) - (1 - \Phi(0.04166...))$ $= 2 \times \Phi(0.04166...) - 1$ $= 2 \times 0.5167 - 1$ $= 0.0334$	B1	For one correct continuity correction used	
		i		M1	For standardizing	
		i		M1	For correctly structured probability calculation.	
		i		A1	CAO inc use of diff tables. Allow 0.0330 – 0.0340 www. <u>Examiner's Comments</u> Most candidates obtained a correct answer. A small but significant number did not use one or both of the continuity corrections. Most used the difference column of the Standard Normal table correctly to provide suitably accurate answers. A relatively small number struggled with the structure of the calculation.	
		ii	$P(Y \geq 80) = P\left(Z \geq \frac{79.5 - 76}{12}\right)$ $= P(Z > 0.2917) = 1 - \Phi(0.2917)$ $= 1 - 0.6148 = 0.3852 = 0.385 \text{ to 3 sig fig}$	B1	For correct cc used	
		ii		M1	For correct structure	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii		A1	CAO do not allow 0.386 <u>Examiner's Comments</u> Many candidates answered correctly, but a common mistake was to omit or to provide an incorrect continuity correction. A relatively small number did not use the difference column correctly.	
		iii	$3 \times 0.3852 \times 0.6148^2 = 0.4368$	M1	$3 \times \text{their } p \times (1 - \text{their } p)^2$	
		iii		A1	FT their p . Allow 2sf if working seen. <u>Examiner's Comments</u> Most candidates knew and applied the method correctly but many were dependent on the FT to gain the 2 marks. A small number omitted the x3 from their binomial calculation.	
		iv	EITHER: $P(\text{Score} \geq k) = 0.1$			
		iv	$\Phi^{-1}(0.9) = 1.282$	B1	For 1.282	
		iv		M1	Allow $k - 0.5$ used for k . Positive z used.	
		iv	$k = 76 + (1.282 \times 12) = 91.38$ or $k = 76 + 0.5 + (1.282 \times 12) = 91.88$	A1	For 91.38 or 91.88	
		iv	$91.38 > 90.5$ or $91.88 > 91$	M1	Relevant comparison (e.g. diagram)	www
		iv	so lowest reported mark = 92	A1		
		iv	OR Trial and improvement method	M1	M1 for attempt to find $P(\text{Mark} \geq \text{integer})$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$P(\text{Mark} \geq 91) = P(\text{Score} \geq 90.5) = 0.1135$	A1	A1 for 0.1135	
		iv	$P(\text{Mark} \geq 92) = P(\text{Score} \geq 91.5) = 0.0982$			
		iv	$P(\text{Mark} \geq 91) > 10\%$ and $P(\text{Mark} \geq 92) < 10\%$	M1	M1 for comparisons	www
					<u>Examiner's Comments</u> 1.282 was identified by the majority of candidates who went on to set up a correct equation and arrive at 91.38 or 91.88. Many of these gave 92% as the final answer but many others gave 91%. Others rearranged incorrectly and arrived at 60.6 for the first calculation. Few candidates demonstrated a proper understanding of the requirement of this question.	
		iv	so lowest reported mark = 92	A1		
		v	$P(Y \leq 50) = 0.2$			
		v		B1	For 50.5 used	
		v		B1	For -0.8416. Condone - 0.842	
					Condone 0.8416 if numerator reversed.	
		v		M1	For structure.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	$\mu = 50.5 + (12 \times 0.8416) = 60.6$	A1	CAO <u>Examiner's Comments</u> In this part, the continuity correction was omitted, or an incorrect value was used, by many candidates. Many used +0.8416 leading to 60.0992 which was a common answer. The issue of over-specification was most apparent in this part of the question.	
			Total	18		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7		i		G1	For suitably labelled axes. Condone absence of scale here.	
		i		G2,1,0	G2 for 11 points correctly plotted relative to a suitable linear scale. G1 if 9 or 10 correctly plotted. G0 if 3 or more incorrectly plotted/omitted Allow axes interchanged Examiner's Comments The scatter diagram was well drawn by many, with some choosing more manageable scales than others.	
		ii	(The points in the scatter diagram) do not appear to be roughly elliptical.	E1	For "not elliptical".	

Question			Answer/Indicative content	Marks	Part marks and guidance																																																																									
		ii	The population may not have a bivariate Normal distribution.	E1	For not underlying bivariate Normal. Do not allow “the data” in place of population/underlying. Allow “data is not from a bivariate Normal distribution”. Do not allow Normal bivariate.... Examiner's Comments The key points concerning the absence of any discernible elliptical shape and the corresponding questioning of the underlying bivariate Normal population were handled well by many. Though most candidates managed to comment on the lack of an elliptical spread of points, often with poor spelling of ‘ellipse’, there were still many candidates who struggled to differentiate between data and population.																																																																									
		iii	<table><tr><td>Percentage</td><td>33</td><td>6</td><td>58</td><td>35</td><td>81</td><td>69</td><td>61</td><td>7</td><td>74</td><td>71</td><td>17</td></tr><tr><td>Fertiliser use</td><td>76</td><td>44</td><td>6</td><td>68</td><td>3</td><td>10</td><td>7</td><td>176</td><td>5</td><td>137</td><td>157</td></tr><tr><td>Rank percentage</td><td>4</td><td>1</td><td>6</td><td>5</td><td>11</td><td>8</td><td>7</td><td>2</td><td>10</td><td>9</td><td>3</td></tr><tr><td>Rank Fertiliser</td><td>8</td><td>6</td><td>3</td><td>7</td><td>1</td><td>5</td><td>4</td><td>11</td><td>2</td><td>9</td><td>10</td></tr><tr><td>d</td><td>4</td><td>5</td><td>-3</td><td>2</td><td>-10</td><td>-3</td><td>-3</td><td>9</td><td>-8</td><td>0</td><td>7</td></tr><tr><td>d²</td><td>16</td><td>25</td><td>9</td><td>4</td><td>100</td><td>9</td><td>9</td><td>81</td><td>64</td><td>0</td><td>49</td></tr></table>	Percentage	33	6	58	35	81	69	61	7	74	71	17	Fertiliser use	76	44	6	68	3	10	7	176	5	137	157	Rank percentage	4	1	6	5	11	8	7	2	10	9	3	Rank Fertiliser	8	6	3	7	1	5	4	11	2	9	10	d	4	5	-3	2	-10	-3	-3	9	-8	0	7	d ²	16	25	9	4	100	9	9	81	64	0	49	M1	For ranking (allow ranks reversed)	
Percentage	33	6	58	35	81	69	61	7	74	71	17																																																																			
Fertiliser use	76	44	6	68	3	10	7	176	5	137	157																																																																			
Rank percentage	4	1	6	5	11	8	7	2	10	9	3																																																																			
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d ²	16	25	9	4	100	9	9	81	64	0	49																																																																			
		iii		M1	For d^2																																																																									
		iii	$\Sigma d^2 = 366$	A1	For Σd^2 (May be embedded in the calculation)																																																																									
		iii	$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 -$ $\frac{6 \times 366}{11 \times 120} = 1 - \frac{2196}{1320}$ $= 1 - 1.6636$	M1	For method for r_s																																																																									

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= -0.664$ (to 3 s.f.) [<i>allow</i> -0.66 to 2 s.f. or $-73/110$]	A1	FT their Σd^2 provided $-1 < r_s < 0$, and ranking used. NB No ranking scores 0/5 Examiner's Comments Many achieved full marks here. Only a few candidates reversed the ranking of one of the sets of values. Errors tended to involve mistakes in ranking, in adding the squares of the differences or in rounding the final answer. Very few candidates failed to rank the data.	
		iv	H_0 : no association between percentage of population living in rural areas and fertiliser use (in the population of countries)	B1	For null hypothesis in context NB H_0 H_1 not ρ .	
		iv	H_1 : negative association between percentage of population living in rural areas and fertiliser use (in the population of countries)	B1	For alternative hypothesis in context. Context needed in at least one of the hypotheses.	
		iv		B1	For population of countries or underlying population.	
		iv	One tail test critical value at 1% level is -0.7091	B1	For ± 0.7091 No further marks from here if incorrect.	
		iv	Since $-0.664 > -0.7091$ [or $0.664 < 0.7091$] there is ...	M1	For sensible comparison of their " -0.664 " with ± 0.7091 seen , leading to conclusion, only if $-1 < \text{their } r_s < 0$.	

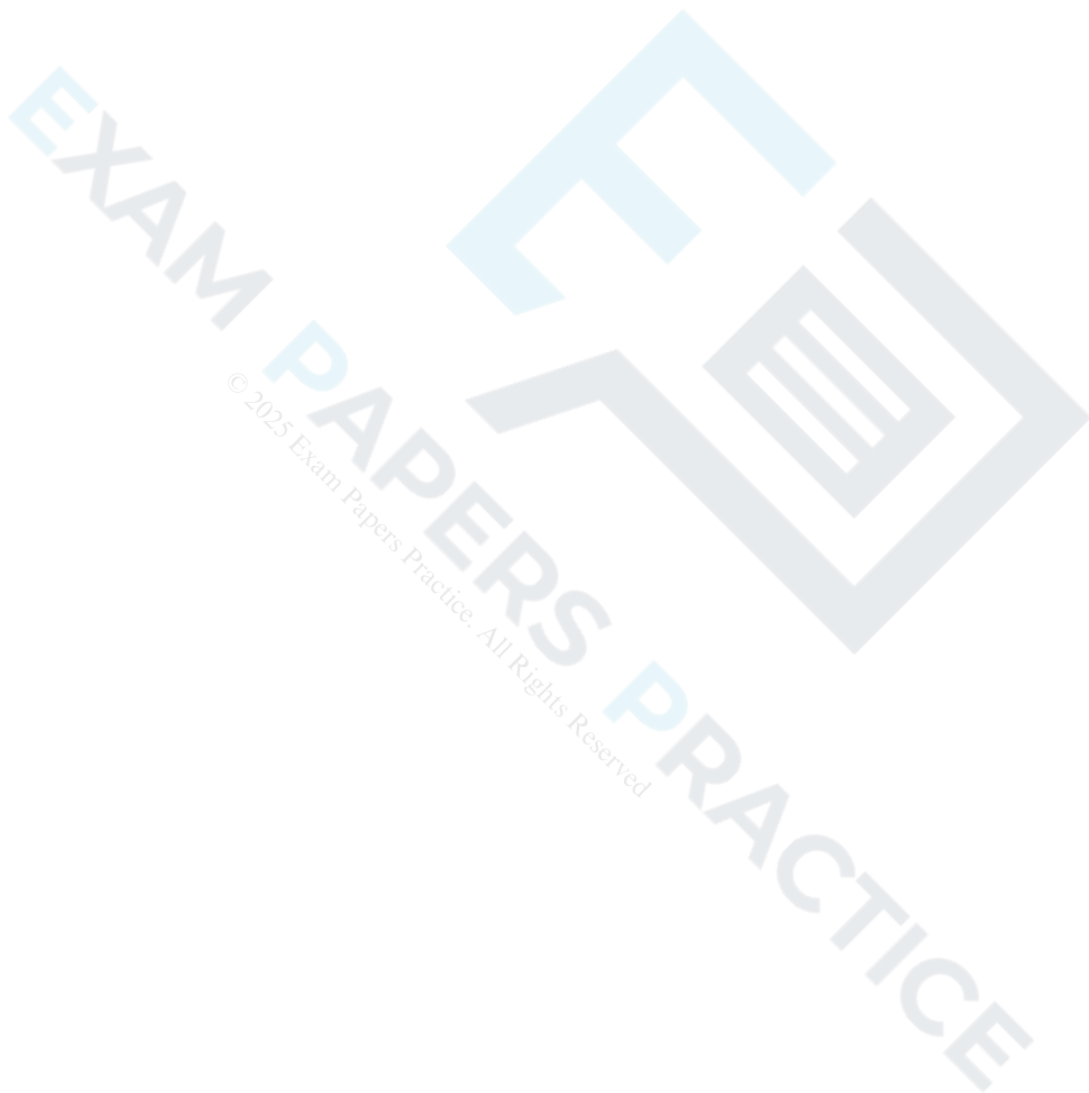
Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	...insufficient evidence to reject H_0 . There is insufficient evidence to suggest that there is negative association between percentage of population living in rural areas and fertiliser use (in the population of countries)	A1	for not significant, oe, and correct conclusion in context. FT their r_s with correct cv. Examiner's Comments This part proved to be challenging for many candidates, especially in the identification of the underlying population of countries involved. Inappropriately worded hypotheses, using 'correlation' in place of 'association', were seen in a few cases. Many realised the one-sided nature of the test which investigated the possible negative association and went on to use the appropriate critical value to obtain a suitable conclusion in context. Few candidates expressed the conclusion in terms of the null hypothesis.	
		v	It means that the probability of rejecting H_0 given that it is correct is 1% o.e.	E1	Allow "the probability of a false positive is 1%", "the probability of a Type I Error is 1%". Do not allow "It means that the probability rejecting H_0 when it should have been accepted is 1%" Examiner's Comments There were many correct answers to this part of the question, either in terms of incorrectly rejecting H_0 or one of the equivalents. Many answers referring to 'accuracy' or 'reliability' were seen.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		vi	None	E1	Examiner's Comments	
					Few candidates appeared to understand that for this test no modelling assumptions about the underlying distribution are required.	
			Total	18		
8		a	$0.758 > 0.279$ So there is sufficient evidence of correlation (in the population)	M1(AO1. 1) A1(AO2. 2b) [2]	Oe but not evidence of positive correlation.	
		b	E.g. diagram shows positive correlation overall, but the data consists of two distinct clusters. E.g. neither of the two clusters show evidence of correlation	B1(AO2. 3) B1(AO2. 2b) [2]	Accept other suitable correct comments	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance		
9		a	Estimated number $= 4 + \frac{16}{3} = 9\frac{1}{3}$ $\frac{9\frac{1}{3}}{80} = 0.1166 \dots$ so proportion is approximately 0.117	M1(AO3.1b) A1(AO1.1) [2]		for attempt at interpolation	
		b	E.g. Midpoints Mean Standard deviation = 3.4	M1(AO1.1) A1(AO1.1) A1(AO1.1) [3]	evidence of valid method for estimation BC Mean in the range 169-171 BC SD in the range 3-3.5		
		c	The histogram e.g. seems to have a rough bell shape e.g. is symmetrical (around the estimated mean) e.g. appears to have all data within 3 s.d. of the mean so this does support the manager's belief	B1(AO3.5a) B1(AO3.5a) [2]	for one reason for at least two reasons and 'supports belief'		
		d	i P(Lifetime > 174) for N(170, 3.4 ²) 0.1197	M1(AO3.4) A1(AO1.1)	oe BC FT their mean and standard deviation		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	ii Answer is very similar to estimate in part (a)	B1(AO3.5a)		
				[3]		
		e	Either Test statistic $= \frac{207.3 - 210}{3.4 / \sqrt{8}} = -2.246$ Lower 5% level 1 tailed critical value of $z = -1.645$ $-2.246 < -1.645$ so significant or under $H_0, \bar{X} \sim N\left(210, \frac{3.4^2}{8}\right)$ $P(\bar{X} \leq 207.3) = 0.01235$ $0.01235 < 0.05$ so significant There is sufficient evidence to reject H_0 There is sufficient evidence to conclude that the mean lifetime is less than 210 minutes.	M1(AO3.4) A1(AO1.1) B1(AO1.1) M1(AO3.4) A1(AO1.1) B1(AO1.1) A1(AO2.2b) E1(AO2.4) [5]	Must include $\sqrt{8}$ For comparison leading to correct conclusion BC	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	15	


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	= 0.5654	1)	$0.92^{29} \times$
		[2]	$0.08^1 +$
			0.92^{30}
			BC

Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	Let p = probability that a train arrives on time $H_0: p = 0.92$ $H_1: p < 0.92$ Let $X \sim B(18, 0.92)$ $P(X \leq 13) = 0.0116$ [$> 1\%$] $P(X \leq 12) = 0.0021$ [$< 1\%$] The critical region is $X \leq 12$	B1(AO2.5) B1(AO1.1) M1(AO1.1) M1(AO1.1) A1(AO2.2a) [5]	For definition of p For H_0 and H_1 For probability $P(X \leq \text{any whole number value 1 to 18})$, Both $P(X \leq 13)$ and $P(X \leq 12)$ For correct critical region stated	Allow FT from $H_1: p < 0.92$ OR $H_1: p \neq 0.92$	
			Total	9			

Question			Answer/Indicative content	Marks	Part marks and guidance		
11			$H_0 : p = 0.49$ $H_1 : p \neq 0.49$ p is the probability that a voter selected at random supports Mr Evans X is the number of voters who support Mr. Evans. Under $H_0 X \sim B(38, 0.49)$ $p(X \leq 13) = 0.047(46439\dots)$ $0.047 > 0.025$ Not significant There is insufficient evidence at the 5% level to suggest that support for Mr Evans has changed.	B1(AO1.1) B1(AO1.1) B1(AO2.5) B1(AO1.1) M1(AO1.1) A1(AO2.2b) E1(AO2.4) [7]	DR BC Compares their 0.047 with 0.025 Or reject H_0 Conclusion in context		
			Total	7			
12		a	Negative skew	B1(AO1.2) [1]			
		b	[They are all less than 0.00005] and the table rounds values [to 4 decimal places]	E1(AO2.4) [1]	Allow equivalent explanation in words		
		c	(18)	B1(AO1.1) [1]			
		d	(0)	1(AO2.2a) [1]			
			Total	4			

Question			Answer/Indicative content	Marks	Part marks and guidance		
13		a	$H_0p = 0.2$ $H_1p < 0.2$ p is the proportion of male adult pilot whales $X \sim B(43, 0.2)$ $P(X \leq 3) = 0.0178$ BC $0.0178 < 0.05$ so result is significant There is sufficient evidence to suggest at the 5% level that the proportion of males in the population of adult pilot whales is less than 20%	B1(AO 1.1) B1(AO 2.5) M1(AO 3.3) B1(AO 3.4) M1(AO 1.1) A1(AO 2.2b) [6]	For both hypotheses Definition of p soi Comparison of their p -value with 0.05		
		b	Sample mean is 3.09 Sample variance is 0.295 $H_0 : \mu = 2.98$ $H_1 : \mu \neq 2.98$ μ is the population mean length of female pilot whales $\bar{X} \sim N(2.98, \frac{0.295}{39})$ $p(\bar{X} \geq 3.09) = 0.1029$ $0.1029 > 0.025$ so result not significant There is insufficient evidence at the 5% level to suggest that the mean length of female pilot whales has changed	B1(AO 1.1) B1(AO 1.1) B1(AO 1.1) B1(AO 2.5) M1(AO 3.3) B1(AO 3.4) M1(AO 1.1) A1(AO 2.2b) [8]	soi For both hypotheses soi Comparison of their p -value with 0.025 BC	NB 3.0948 7179487 and 0.2947 09851552	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	14	


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Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i	$(X \sim B(12, 0.75))$ A) $P(X=9) = \binom{12}{9} \times 0.75^9 \times 0.25^3 = 0.258 \text{ (0.258103...)}$	M1	For $0.75^9 \times 0.25^3$	
		i	Or: From tables $P(X \leq 9) - P(X \leq 8) =$ $0.6093 - 0.3512 =$ 0.2581	M1 M1 A1 M2 A1 [3]	For $\binom{12}{9} \times p^9 \times q^3$ With $p + q = 1$ Also for $220 \times 0.00117...$ Allow 0.26 or better with working CAO For 0.6093 – 0.3512 CAO Examiner's Comments Around 90% of candidates gained full marks here, with most using the formula, rather than tables.	
		i	B) $1 - 0.3512 = 0.6488$	M1 A1 [2]	For 0.3512 CAO Accept 0.649 and 0.65 with working For $P(X=9) + P(X=10) + P(X=11) + P(X=12)$ allow M1A1 for awrt 0.649. Otherwise M0A0.	
					Examiner's Comments Again this was well answered, usually by use of cumulative probability tables, although some candidates did calculate the four probabilities, usually summing them correctly. A few candidates forgot to subtract from 1, and a few just subtracted $P(X=8)$ from 1 rather than $P(X \leq 8)$.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	(Let $X \sim B(18, 0.75)$) Let p = probability of dog having allergy relieved by the new shampoo (for population) $H_0: p = 0.75$ $H_1: p > 0.75$ H_1 has this form as the test is to determine whether the new shampoo relieves the symptoms of a higher proportion of dogs who suffer from the allergy. For use of $B(18, 0.25)$, please consult your Team Leader	B1 B1 B1 E1 [4]	For definition of p (in context) Do NOT allow <u>number</u> in place of probability. See below for additional notes For H_0 For H_1 Dep on > 0.75 used in H_1 E0 for simply stating H_1 in words Condone number instead of proportion. Do Not allow just 'proportion will be higher' or similar. Examiner's Comments Candidates did well on this part, with over 80% gaining at least 3 marks out of 4. Most candidates scored the first two marks for the hypotheses, with many knowing that they needed to define p, thus scoring the third mark, although some definitions were wrong. For example 'p = the probability that dogs suffer from the allergy'. A valid explanation of the reason for the form of the alternative hypothesis was usually given, even if not always very well worded.	

Question			Answer/Indicative content	Marks	Part marks and guidance
		iii	$P(X \geq 16) = 1 - P(X \leq 15) = 1 - 0.8647 = 0.1353$ $0.1353 > 0.1$ So not significant. Accept H_0 Conclude that there is not enough evidence to support the idea that the new shampoo relieves the symptoms of a higher proportion of dogs who suffer from the allergy. $0.8647 < 0.9$ scores M2 and can get A1 E1 if $P(X \leq 15)$ oe seen and all correct ALTERNATIVE METHOD Provided they are using CR method $P(X \geq 16) = 0.1353$ $P(X \geq 17) = 0.0395$ OR 0.8647 and 0.9605 $0.1353 > 0.1$ or $0.0395 < 0.1$ OR $0.8647 < 0.9$ or $0.9605 > 0.9$ So critical region is $\{17, 18\}$ so not significant. or 16 not in CR so not significant Conclude that there is enough evidence to support the idea that the new shampoo relieves the symptoms of a higher proportion of dogs who suffer from the allergy.	M1* *M1dep A1* E1dep [4] B1 M1 A1* E1*dep	For sight of 0.1353 or 0.135 For (explicit) comparison with 10% or 0.1 Do NOT FT wrong H_1 but first mark available if H_1 or H_0 wrong For A1 need $P(X \geq 16)$ somewhere oe eg $P(\geq 16)$ Allow SC2 for clearly indicating use of $B(18, 0.75)$ but with no mention of 0.1353 with convincing reasoning and final answer correct No marks if point probabilities used. Do not condone number instead of proportion Must include 'not enough evidence' oe For both probabilities Do not insist on correct notation as candidates have to work out two probabilities for full marks. For at least one comparison with 10% Allow comparison in form of statement 'critical region at 10% level is ...' CAO dep on the two correct probabilities Ignore any work on lower critical region No marks if CR not justified. However SC2 above still applies Condone $X \geq 17$, , oe but not $P(X \geq 17)$ etc Assume using first method unless you are convinced that candidate is using CR method. No marks if point probabilities used

Question			Answer/Indicative content	Marks	Part marks and guidance	
					Examiner's Comments Approximately half of the candidates scored full marks in this part and also the final part. Most candidates started off correctly by using 0.1353, but there were still quite a few who used point probabilities, scoring zero. The use of 0.0395 was not uncommon, again scoring zero. There were a few candidates who did not compare their probability to the significance level and so could only be awarded one mark. Some candidates used the critical region method and in this part the two correct probabilities were used most of the time and compared with the significance level. The final mark was often lost due to failure to provide a statement, failure to include context or failure to include an element of doubt.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$P(X \geq 42) = 1 - P(X \leq 41) = 1 - 0.9084 = 0.0916$ $0.0916 < 0.1$ or $0.9084 > 0.9$ So significant. Reject H_0 Conclude that there is enough evidence to support the idea that the new shampoo relieves the symptoms of a higher proportion of dogs who suffer from the allergy.	B1 M1* A1* E1*dep [4]	For use of $P(X \leq 41)$ For comparison with 10% dep on first two marks NB If more than one attempt please mark the final one. Do not penalise 'number' rather than 'proportion' twice in parts A and B NB No marks for critical region method unless find $P(X \leq 40) = 0.9084 - 0.0721 = 0.8363$ in which case follow above scheme for part (iii)(A) so should have $0.1637 > 0.1$ and $0.0916 < 0.1$ or $0.8363 < 0.9$ and $0.9084 > 0.9$ etc (giving CR{42, 43, 44, 45, 46, 47, 48, 49, 50}) Examiner's Comments The majority of candidates used 0.0916 but there again there were quite a lot of candidates who used point probabilities. Some candidates used a critical region method but there were far too many who didn't use the correct two cumulative probabilities, but just 0.0916 and 0.0453, comparing both to 0.1. To score marks using the critical region method candidates needed to compare both 0.0916 and 0.1637 to 0.1 to justify the critical region.	
			Total	17		

Question			Answer/Indicative content	Marks	Part marks and guidance	
15		a	$= 6.8 + 1.5 \times (6.8 - 2.5) [= 13.25]$	B1(AO3.1b) [1]		
		b	Streng – large sample size th Weakn– sample is not ess random – removal of outliers may not be justifiable	B1(AO2.2a) B1(AO2.3) B1(AO2.4) [3]		
		c	eg pmcc may not be appropriate as scatter does not appear to be linear eg sample not random so calculating a correlation coefficient may not be valid eg correlation and causation are not the same thing eg the results of a hypothesis test are only suggestive, not conclusive (although highly suggestive in this case)	E1(AO2.4) E1(AO2.4) E1(AO2.4) [3]	Allow 1 mark for each distinct (sensible) reason up to a maximum of 3 	
			Total	1		

Question			Answer/Indicative content	Marks	Part marks and guidance		
16		a	eg some people have more than one phone oe	E1(AO2.4) [1]	Some of the other figures in the LDS are greater than 1		
		b	E.g. data missing E.g. removal of outlier data items	B1(AO1.1b) B1(AO1.1b) [2]			
		c	Upper quartile	B1(AO1.2) [1]			
		d	$X \sim N(996, 407^2)$, $p = 0.75$ 1271	M1(AO3.4) A1(AO1.1b) [2]	BC allow awrt 1300	if M0 allow B2 for awrt 1300 unsupported	
		e	p -value = 0.127 > 0.05 not significant Insufficient evidence to suggest that the population mean of the number of mobile phones/1000 population has increased	M1(AO1.1b) M1(AO1.1b) A1(AO2.2b) E1(AO2.2b) [4]	from screenshot comparison with 0.05 FT		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		f	Sample size of 12 is small compared to the number of countries in the population so inference may be unreliable	B1(AO2.3) [1]	Any sensible comment on the premise that 12 is a relatively small sample size		
			Total	1			

Question			Answer/Indicative content	Marks	Part marks and guidance		
17		a	9×22.6 or 11×22.8 or 17×23.7 soi 203.4 , 250.8 and 402.9 isw $8 \times 8.9 + 9 \times 22.6^2$ or $10 \times 10.618 + 11 \times 22.8^2$ or $16 \times 16.9925 + 17 \times 23.7^2$ soi 4668.04 , 5824.42 and 9820.61 isw	M1(AO3.1a) A1(AO1.1b) M1(AO2.1) A1(AO1.1b) [4]	if M0 allow SC1 for use of n instead of $n - 1$ to find all three $\sum x^2$ accept answers to 4 sf or more		
		b	23.16486 to 3 sf or more $12.7(35\dots)$ to 3 sf or more	B1(AO1.1b) B1(AO1.1b) [2]	from $\frac{203.4+250.8+402.9}{37}$ from $\frac{4668.04+5824.42+9820.61-37 \times 23.1648}{36}$		
		c	Use of $N(26.1, \frac{12.735}{37})$ $26.1 \pm 1.96 \times \sqrt{\frac{12.735}{37}}$ Critical region: mean < 24.95 and mean > 27.25	M1(AO3.3) M1(AO3.1a) A1(AO3.4) [3]	FT their variance of combined sample allow use of $z = 1.645$ for M1		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	The evidence suggests that the mean mass of adult voles has changed since the sample mean is inside the critical (rejection) region	E1(AO2.2b) [1]	or 'outside acceptance region'		
			Total	4			
18			p is the probability that a cutting treated with Miracleroot develops into a healthy new tree $H_0: p = 0.8$ $H_1: p < 0.8$ Let $X \sim B(500, 0.8)$ $P(X < 380) = 0.01609$ <i>their</i> $0.01609 > 0.01$ so accept H_0 no evidence to suggest that manufacturer's claim is optimistic	B1(AO2.5) B1(AO1.1) B1(AO3.3) M1(AO3.4) M1(AO1.1) E1(AO2.2b) E1(AO3.2a) [7]	For definition of p For H_0 and H_1 BC		
			Total	7			

Question		Answer/Indicative content	Marks	Part marks and guidance		
19		$H_0: p = 0.62$ $H_1: p > 0.62$ p is the proportion of adults over 65 in the (UK population) who use the online social media platform	B1(AO1.1) B1(AO1.1) B1\$(AO2.5) B1&(AO1.1) M1&(AO1.1)	Allow null Allow alternative May be seen in hypotheses Allow probability		
		$1 - P(X \leq 45) = 0.0068(1)$ $0.0068 < 0.01$ Result is significant or "reject H_0 " The evidence suggests that the proportion of adults over 65 (in the UK population) using platform has increased from 62%	A1(AO2.2b) E1(AO2.4) [7]	NB from use of $B(59, 0.62)$ Comparison of their 0.0068 with 0.01 or 0.68% with 1%; not allowed from point probability Depends on $B1 \& M1 \&$ Conclusion in context Depends on all other marks except $B1\$$	Allow for sight of 0.9932, (from 0.99318902) Allow 'accept H_1 ' OR Critical region ≥ 46 $B1$ 46 in critical region, $M1$, hence conclusion	
				Examiner's Comments The majority of candidates made valid attempts at this question. Common errors at the start of the question included defining H_1 as $p <$		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					0.62 and failing to define p. Some candidates wrongly used the probability that $X \leq 46$ in the test, others wrongly used a point probability. Most candidates who arrived at the correct p value correctly compared it with 0.01 and went on to reject H_0. A small number of otherwise successful candidates did not give a conclusion in the context of the question. There were a small number of scripts from candidates who either omitted the question completely, or wrote vague comments such as 46 out of 59 is about 78%, which is a lot more than 62%.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance		
20		a	calculation of $P(X < 14)$ and $P(X > 18)$ 0.3085 and 0.0668 to 1 sf or better these figures do not support the model	M1 (AO 3.4) A1 (AO 1.1) A1 (AO 3.5a) [3]	or solves $-1.476 = \frac{14 - \mu}{\sigma}$ an $0.496 = \frac{18 - \mu}{\sigma}$ d simultaneously $\mu \approx$ and σ $17 \approx 12.02$ 17 is (relatively) far from 15 so not a good fit the second A1 is only available if the first A1 is awarded allow SC2 for showing the model is not a good fit for either value with all working correct or for a complete argument based on symmetry which refers to both tails	or solves $-1.476 = \frac{x - 15}{2}$ an $0.496 = \frac{x - 15}{2}$ d $x = 12.048$ and 15.992 to nearest whole number or better which are not close to 14 and 18 $\frac{14 - 15}{2}$ and r $\frac{18 - 15}{2}$ evaluated – 0.5 and 1.5 obtained which are not close to –1.476 and 0.496 respectively	
					<u>Examiner's Comments</u>		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					Candidates who did well in this question made appropriate calculations, showed full working and explained the significance of their results. Candidates who did less well gave incomplete reasoning and/or made slips in their calculations.	
		b	$\Phi^{-1}(0.07) = -1.476 = \frac{14-\mu}{2}$ $[\mu = 16.95]$ OR $\Phi^{-1}(0.69) = 0.496 = \frac{18-\mu}{2}$ $[\mu = 17.008]$ $[\mu =]17$	M1 (AO 3.5c) A1 (AO 2.4) [2]	<i>alternative y</i> since the variance is assumed to be correct, the mean must be as far above the midpoint as it was previously below it. 16 + 1 = 17 if M0 allow B2 for 17 unsupported	
					Examiner's Comments Candidates who did less well in this question ignored the request for an answer correct to two significant figures.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$z = \pm 1.96$ used $\frac{16 - \mu}{\frac{\sigma}{\sqrt{n}}} < -1.96$ or $\frac{\mu - 16}{\frac{\sigma}{\sqrt{n}}} > 1.96$ $\sqrt{n}$ isolated $\frac{16 - \mu}{\frac{\sigma}{\sqrt{n}}} < -1.96$ oe from their $[n >] 15.3664 - 15.4$ $n = 16$ cao	B1 (AO 1.1a) M1 (AO 3.1b) M1 (AO 2.1) A1 (AO 3.4) A1 (AO 2.2b) [5]	allow method marks only if other z – value, eg – 1.645 used; FT μ eg $\sqrt{n} > 2 \times 1.96$ previous A1 must be awarded for the award of final A1	NB 1.9599 63985... rounded to 3 or more sf M0 if other value for σ used all marks are available if works with = instead of < or > throughout, but withhold final A1 if works with < instead of > or > instead of < throughout	Examiner's Comments Candidates who did well in this question recognised that the hypothesis test was two-tailed and worked with $\sqrt{n}$ in their calculations. Candidates who did less well worked with an incorrect z value or an incorrect form of the standard deviation. Exemplar 2 $P(X < 14) = 0.07$ $P(X > 15) = 0.11$ $\frac{14 - \mu}{\frac{\sigma}{\sqrt{n}}} = -1.67579$ $\frac{15 - \mu}{\frac{\sigma}{\sqrt{n}}} = 1.3$ $\frac{14 - 15}{\frac{\sigma}{\sqrt{n}}} = -0.5$ $\frac{15 - 15}{\frac{\sigma}{\sqrt{n}}} = 0$ So our model is not a good fit as the z values are far from the actual probabilities. $\frac{14 - \mu}{\frac{\sigma}{\sqrt{n}}} = -1.67579$ $-\mu = -2.95176 - 14$ $\mu = 16.95 = 17$ $\mu = 17$

Question			Answer/Indicative content	Marks	Part marks and guidance
					$\mu \sim N(17, 2^2)$ $H_0: \mu = 17 \quad H_1: \mu \neq 17$ where μ is mean journey time. $n = 16 < 17$ so you want smallest value, find lower tail $\alpha = 0.05$ z value of 0.05 is 1.645 $\frac{16 - 17}{\frac{2}{\sqrt{16}}} = -1.645$ $\frac{16 - 17}{\frac{2}{\sqrt{16}}} \leq \frac{z}{\frac{2}{\sqrt{16}}} \Rightarrow n \leq \left(\frac{z}{0.607956} \right)^2$ $n \leq 10.82$ $n \geq 0$ or 11 is too large $n = 10$ Note that in part (a) the candidate has chosen a less frequently seen approach, identified in the RH column of the mark scheme. Part (b) is well done. In part (c) the wrong choice of z proved costly.
			Total	10	

Question			Answer/Indicative content	Marks	Part marks and guidance	
21		a	(i) the cumulative frequencies have been plotted against the mid-points of the class intervals, mis-plotting [at centre of each class] reduces estimate (by 2.5) oe	B1 (AO 2.4) B1 (AO 2.4) [2]	<u>Examiner's Comments</u> Candidates who did well in this question recognised that the cumulative frequencies had been plotted at the mid-point of the intervals instead of at the upper limit.	
			(ii) grouped data has been used grouping has slightly reduced the error introduced by misplotting (because the error is less than 2.5)	B1 (AO 2.4) B1 (AO 2.4) [2]		

Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	percentage unemployment is often estimated oe	E1 (AO 2.4) [1]	allow data (on percentage unemployment) is not available for all countries in Europe oe <u>Examiner's Comments</u> Candidates who did less well based comments on general geographical or economic ideas, rather than specifically related to issues related to the estimation of median values.		
		c	there are many other countries in the pre-release material; it is very unlikely that a random sample would only include European countries.	E1 (AO 2.4) [1]	<u>Examiner's Comments</u> Candidates who did well on part (b) and part (c) were familiar with the pre-release material made appropriate comments.		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		d	negative correlation / association (may be embedded) comparison of p -value with 0.05 or 0.01 or other appropriate significance level and supporting comment	B1 (AO 2.2b) B1 (AO 2.2b) [2]	if B0B0 allow SC2 for eg comment on no significant association justified by comparison of p-value with appropriate significance level (eg 0.025) <u>Examiner's Comments</u> Candidates who did well commented on the nature of the correlation. They compared the p-value with a significance level and then made an appropriate deduction. Candidates who did less well compared the correlation coefficient with the p-value or did not comment on the association at all.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		e	(even though this is interpolation), the scatter / weak correlation / presence of an outlier would suggest that the use of a line of best fit is inappropriate	E1 (AO 2.2b) [2]	allow explanation based on the value for Kosovo being an outlier or on it lying in the (large) gap in the scatter <u>Examiner's Comments</u> Candidates who did well commented on the nature of the scatter or the position of 35.3 relative to the given values to justify their comment.	
			Total	9		

Question		Answer/Indicative content	Marks	Part marks and guidance		
22		$H_0: \mu = 56$ $H_1: \mu > 56$ μ is the population mean time taken by Statto to complete his walk Use of $N(56, \frac{4^2}{19})$ 0.9459 BC $1 - 0.9459 > 0.05$ No evidence to reject H_0 at 5% level oe There is no evidence to suggest at the 5% level that the mean time taken by Statto to complete his walk has increased	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO1.1) M1 (AO3.4) A1 (AO1.1) E1 (AO2.2b) [7]	Both hypotheses		
		Total	7			

Question			Answer/Indicative content	Marks	Part marks and guidance		
23		a	0.6	B1 (AO1.1) [1]	BC		
		b	each egg has the same (constant) probability of being cracked	E1 (AO2.4) [1]	OR the probability of any particular egg being cracked is independent of any other egg being cracked or		
		c	$np = \text{their } 0.6$ $p = 0.1$	M1 (AO3.1b) A1 (AO1.1) [2]			
		d	159.4, 106.3, 29.5, 4.4, 0.4, 0.0, 0.0	M1 (AO3.4) M1 (AO1.1) A1 (AO1.1) [3]	Use of B(6, 0.1) so Multiplication by 300 All correct	If unsupported, allow B3 for all frequencies correct, B2 if one error, B1 if two errors	
		e	The theoretical frequencies are close to the observed values, so Rose's model is a good fit.	B1 (AO3.2b) [1]			

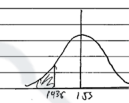
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Question		Answer/Indicative content	Marks	Part marks and guidance		
24		$\bar{X} = 130$ seen $H_0 : \mu = 142$ $H_1 : \mu < 142$ μ is the population mean length of time of meetings use $N\left(142, \frac{17.4^2}{9}\right)$ of $P(\bar{X} < 130) = 0.01927...$ $0.01927... < 0.05$ So [reject H_0], there is evidence at the 5% level to suggest that the mean length of meeting time has decreased.	B1 (AO 1.1) B1(AO 1.1) B1(AO 2.5) M1(AO 3.3) A1(AO 1.1) M1(AO 3.4) A1(AO 2.2b) [1]	BC soi BC FT comparison of their probability		
		Total	7			

Question			Answer/Indicative content	Marks	Part marks and guidance	
25	a		[=]153	B1 (AO1.1) [1]	<u>Examiner's Comments</u> Most candidates did well in this question and simply wrote down the correct answer. Candidates who did not do well usually wrote the answer 168.	
	b		$z = \pm 1.645$ used their positive $1.645 = \frac{183 - \text{their } 153}{\sigma}$ oe (= 18.237...to 18.248...) $\sigma = 18.2$ cao	B1 (AO1.1) M1 (AO2.1) A1 (AO1.1) [3]	or $\pm 1.644(85..)$) M0 if continuity correction used	<u>Examiner's Comments</u> Candidates who did well in this question used the standard Normal distribution in conjunction with the given information to obtain a value for the standard deviation, giving their answer to the specified precision. Candidates who did less well worked with either an incorrect z-score or an incorrect mean. Sometimes they gave their answer to a different precision.

Question		Answer/Indicative content	Marks	Part marks and guidance		
	c	$\left[\frac{16}{452} = \right] 0.035(398 \dots)$ their $P(X < 120)$ from $N(\text{their } 153, \text{their } 18.2^2)$ probability of 0.0349 to 0.0352 which agrees to 2 sf	B1 (AO3.1b) M1 (AO1.1) A1 (AO2.2b) [3]	allow percentage M0 if continuity correction used allow percentage ----- --- or B1 for their $P(X < 120)$ then M1 for 452 $\times$ their 0.03490.. then A1 for 15.77 to 15.91 which is close to 16 oe ----- --- or B1 as main scheme then M1 for $\sigma = \frac{120 - 153}{\text{their } -1.809}$ and 18.3 $\approx$ 18.2 for A1	or B1 as main scheme then Invnorm(0.0353, their 153, their 18.2) for M1 NB 119.95 to 120.15 which is close to 120 oe for A1 or B1 as main scheme then $z = \frac{120 - \text{their } 153}{\text{their } 18.2}$ for M1 NB -1.809 to -1.813 and Invnorm(0.0353, 0, 1) to obtain -1.806 to -1.812 which is close to -1.809 to -1.813) for A1	
				Examiner's Comments Candidates who did well in this question used their model to obtain a value for $P(X < 120)$ and then made a sensible comparison with $16/452 = 0.035\dots$ They		

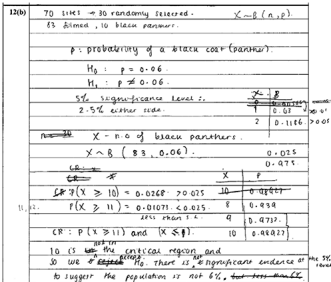
Question			Answer/Indicative content	Marks	Part marks and guidance	
					made efficient use of their calculators.	
					Candidates who did less well made errors in calculating their probabilities, sometimes transposing μ and σ . Sometimes they did not communicate their reasoning clearly.	
	d		$H_0: \mu = \text{their } 153$ $H_1: \mu < \text{their } 153$ μ is the population mean flight time from Magaluf to Liverpool use of $N(\text{their } 153,$ $\frac{\text{their } 18.2^2}{24})$ to find $P(\bar{X} < 143.6)$ awrt 0.0057 to 0.0058.. to 2 or more sf oe isw their 0.0057 correctly compared with 0.01 oe result is significant or reject H_0 or accept H_1 there is sufficient evidence to suggest at the 1% level that the mean flight time from Magaluf to Liverpool is less than 153 / mean flight	B1 (AO1.1) B1 (AO2.5) M1* (AO3.3) A1 (AO1.1) M1dep* (AO3.4) A1 (AO1.1) A1 (AO2.2b) [7]	for both hypotheses ; may be stated in words, but need to see 153 for B1 and population mean for 2 nd B1 or inv Norm(0.01, their 153, $\frac{\text{their } 18.2^2}{24})$ $\bar{X} < 144$ to 144.4 is critical region 143.6 correctly compared with their 144.36 FT their comparison FT their comparison	B0 if other parameter used unless clearly defined as population mean or $z = \frac{143.6 - \text{their } 153}{\frac{\text{their } 18.2}{\sqrt{24}}}$ $z = -2.5248..$ to $-2.5302..$ to 2 or more dp their z compared with -2.326 or -2.33 do not allow eg conclude / prove / indicate or other

Question			Answer/Indicative content	Marks	Part marks and guidance	
			time from Liverpool to Magaluf		assertive statement instead of suggest	
					Examiner's Comments Candidates who did well in this question set out their hypotheses clearly in terms of μ, going on to define it as the population mean flight time from Magaluf to Liverpool. They clearly showed how they achieved their probability, making a clear comparison with the significance level. They went on to give a clear conclusion in context. Candidates who did less well gave incomplete / incorrect definitions and conclusions. Candidates who did badly used an incorrect parameter without defining it clearly and / or used continuity corrections and / or worked with $n = 452$ or $n = 1$. Exemplar 3 $\bar{X} = 143.6$ (if μ be the mean flight time from Magaluf to Liverpool) $n = 94$ $\sigma = 18.2$ $X \sim N(153, 18.2^2)$ $\bar{X} \sim N(153, \frac{18.2^2}{94})$  $H_0: \mu = 153$ $H_1: \mu < 153$ 1% s.c. $P(X < 143.6) = 0.005699$ 4 s.f. $\therefore 0.005699 < 0.01$, there is sufficient evidence to reject H_0, i.e. the mean flight time from Magaluf to Liverpool is less than the mean flight time from Liverpool to Magaluf. The hypotheses are correctly set up but the definition of μ is incomplete,	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					so the second B1 is not earned. The rest of the work is good, apart from the final conclusion, as there is no element of suggestion that the mean flight time from Magaluf to Liverpool may be lower than the mean flight time from Liverpool to Magaluf – it is too assertive.	
	e		reduce the value of μ increase the value of σ or σ^2	B1 (AO3.5c) B1 (AO3.5c) [2]	allow B1 for eg use new sample data to calculate new estimate for σ or σ^2 eg take a bigger sample is insufficient Examiner's Comments Candidates who did well on this question considered how the mean and the standard deviation might be adjusted to improve the model. Candidates who did badly wrote in generalities – “collect more data” was commonly seen.	
			Total	16		

Question		Answer/Indicative content	Marks	Part marks and guidance		
26	a	$0.94^n < k$ or $0.06^n < k$ seen $k = 0.025$ used in inequality as above 60	M1 (AO3.4) B1 (AO1.1) A1 (AO2.2a) [3]	NB 59.617.. or 1.311...to 1 or more dp if unsupported implies M1B1	allow = instead of < 60 unsupported or from trial and improvement scores 3	
Examiner's Comments Candidates who did well in this question worked with 0.94^n and compared it with 0.025. Many achieved a correct answer by trial and improvement instead of the traditional log approach. Candidates who did less well worked with 0.06^n. Some candidates were unable to make any meaningful progress with this question.						
	b	$H_0 : p = 0.06$ allow equivalent in words $H_1 : p \neq 0.06$ p is the probability that a jaguar chosen at random is a black panther / has black coat use of $B(83, 0.06)$ to obtain $P(X \leq K)$ oe $\text{cdfBinomial}(83, 0.06, 9) = 0.973$ to $0.97321...$ or $1 - \text{cdfBinomial}(83, 0.06, 9) = 0.02679$ to 0.027 $1 - \text{their } P(X \leq K)$ compared	B1 (AO1.1) B1 (AO2.5) M1* (AO3.3) A1 (AO1.1) M1dep* (AO3.4) A1 (AO1.1) A1 (AO2.2b) [7]	or p is the proportion of jaguars that are black panthers / have a black coat not $P(X = K)$ or critical region is $X \geq 11$ (ignore lower tail) eg 10	NB $P(X \leq 10) = .98927$. for comparison of their $P(X > K)$ with 0.025 or their $P(X \leq K)$ with 0.975 or stating whether 10 is in their critical	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			with 0.025 or their $P(X \leq K)$ compared with 0.975 oe result is not significant or do not reject H_0 or reject H_1 there is insufficient evidence at the 5% level to suggest that the probability that a jaguar selected at random from this population is a black panther is not 0.06		compared with their critical region oe must have the correct probability or correct critical region for the last two A marks	region allow accept H_0 do not allow eg conclude / prove / indicate or other assertive statement instead of suggest	
					<u>Examiner's Comments</u> Candidates who did well in this question set up the hypotheses correctly and went on to use their calculators efficiently to calculate the required probability. This usually led to a correct comparison with the significance level and a correct conclusion, although this was not always put into context successfully. A few candidates spoiled their answers by being too assertive in their final conclusion. Candidates who did less well gave an incomplete definition of p and/or worked with $P(X > 10)$ instead of $P(X \geq 10)$. Candidates who didn't do well worked with $P(X = 10)$.		

Question			Answer/Indicative content	Marks	Part marks and guidance
					Exemplar 4  The hypotheses are correctly set up but the definition of p is incorrect, so the second B1 is not earned. This candidate opts for the critical region approach. The correct upper tail is identified, and as the lower tail is irrelevant to the conclusion it may be ignored in this case, so the incorrect identification of 1 being in the critical region was not penalised. The rest of the work is good. However, the final conclusion in context is incomplete, and so this response earns 5/7.
			Total	10	

Question		Answer/Indicative content	Marks	Part marks and guidance		
27	a	$\left[\frac{19.8}{36} =\right] 0.55$	B1 (AO1.1) [1]	or $\frac{11}{20}$	Must be dec or simplified frac; but condone 55%	
	b	0.0380(96943116...) BC	B1 (AO1.1) [1]	Correct to at least 2 sf	Allow truncated	
	c	$H_0: p = 0.45$ $H_1: p < 0.45$ p is the probability that a driver speeding / caught speeding / on speed awareness course is female Use or sight of B(25, 0.45) $P(X = 7) = 0.06385(0773831)$ $0.06385 > 0.05$ Result is NOT significant or "accept H_0" There is insufficient evidence (at the 5% level)	B1 (AO1.1) B1 \$(AO2.5) M1 *(AO3.3) A1 (AO1.1) *M1 (AO3.4) B1 (AO1.1) E1 (AO2.2b)	$O H_0: p = R 0.55$ $H_1: p > 0.55$ p is the probability that a driver speeding / caught speeding / on speed awareness course is male OR Use or sight of B(25, 0.55) $P(X = 18) = 1 - P(X =$	Both hypotheses Must not be inconsistent with H_1 Can be implied by $P(X = 7)$ (OR $P(X = 18) = 0.038..$	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			to suggest that drivers speeding / caught speeding / on speed awareness course are more likely to be male. oe	[7]	$17) = 1 - 0.936149 = 0.06385$ FT their probability Dep on 0.06385 Dep all previous marks except B1\$ SC: Use of $p = 0.5$ can score B1B1M1 (can be implied by 0.0216 or 0.0143) A0M1B0E0 May be incorporated in conclusion Not assertive		
					<u>Examiner's Comments</u> Good answers were either based on females, using $B(25, 0.45)$ and the hypotheses $H_0: p = 0.45$ and $H_1: p < 0.45$, or males, using $B(25, 0.55)$ and the hypotheses $H_0: p = 0.55$ and $H_1: p > 0.55$. Most candidates using a correct value of p then went on to compare 0.0639 ($P(X = 7)$ or $P(X = 18)$) with 0.05. Most then correctly stated that the result is not significant, or that H_0 should not be rejected, and then went on to give an appropriate conclusion in context. However, even good candidates often defined p incorrectly as the probability that a female (or male) driver was speeding, rather than the correct		

Question			Answer/Indicative content	Marks	Part marks and guidance	
					definition as the probability that a driver speeding (or caught speeding) was female (or male). Partial credit was allowed for candidates who used the value of 0.5 for p. There were a considerable number of candidates who made little, or no, progress on this question. Exemplar 3 $Let p = \text{proportion of drivers at speed awareness course that are female}$ $H_0: p = 0.45$ $H_1: p < 0.45$ $X \sim B(25, 0.45)$ $P(X \leq 9) = 0.06385$ $As 0.06385 > 0.05$, we accept the H_0. There is not evidence to suggest male drivers are more likely to exceed the speed limit than female drivers. This candidate is set up and concluded the hypothesis test nicely. The conclusion is non-assertive and in context.	

Question			Answer/Indicative content	Marks	Part marks and guidance		
	d		eg Each driver speeding / caught speeding has the same (constant) probability of being selected for the speed awareness course.	E1 (AO2.4) [1]	OR the probability of any particular driver caught speeding being selected for the speed awareness course is independent of any other driver caught speeding being selected.		
					Examiner's Comments This part was often omitted or generic. The context of the question needed to be considered in the answer.		
					Exemplar 4 <u>Events occurred mutually and independently.</u> The candidate has offered up the idea of independence, but has not used this in the context of the question looking at the probability of any one driver caught speeding being selected is independent of any other driver caught speeding being selected.		
			Total	10			

Question			Answer/Indicative content	Marks	Part marks and guidance		
28			$P(X \leq 4) = 0.013(337876\dots)$ or $P(X \leq 3) = 0.004(09062\dots)$ found Hence 4 is not in CR, or 3 is in CR so CR is $\{0, 1, 2, 3\}$ or $X \leq 3$	M1 (AO3.1a) M1 (AO2.2a) A1 (AO3.2a) [3]	Allow M1 for one (correct) of these. Both probabilities must be correct	Accept $P(X \leq 4) > 0.01$ or $P(X \leq 3) < 0.01$	
			Total	3			

Question		Answer/Indicative content	Marks	Part marks and guidance		
29		$H_0: \mu = 5$ oe $H_1: \mu \neq 5$ oe H_1 takes this form as Claud is testing whether the mean length of time is different to 5 oe μ is the population mean time for which computers are kept before being replaced use of $N(5, \frac{2.7054^2}{120})$ to find $P(\bar{X} < 4.8855)$ or $\text{invNorm}(p, 5, \frac{2.7054}{\sqrt{120}})$ where $p = 0.025$ or 0.05; may be implied by 0.3215 or $4.51595\dots$ or $4.59377\dots$ $P(\bar{X} < 4.8855) = \text{awrt } 0.32$ $0.32 > 0.025$ or $4.8855 > 4.5(2)$ not significant or accept H_0 or do not reject H_0 or reject H_1	B1 (AO1.1) B1 (AO2.4) B1 (AO2.5) M1 (AO3.3) A1 (AO1.1) M1 (AO3.4) A1FT (AO1.1) A1FT (AO2.2b) [8]	allow any parameter apart from $\bar{x}$ for population mean as long as clearly defined as (population) mean condone use of 2.6941 (may be rounded) instead of 2.7054 for M1, may be implied by 0.32076 or CR is $\bar{x} < 4.51797$, but A1 not available or CR is $(\bar{X}) < 4.5 - 4.6$ comparison of <i>their</i> probability with 0.025 or comparison of 4.8855 with <i>their</i> critical value from	after B marks M1A1M1A1 A1 may be earned if working with 2.7054 rounded to 2 or more sf $z = \frac{4.8855 - 5}{\frac{2.7054}{\sqrt{120}}}$ For M1 awrt -0.46 A1 (may be implied by -0.466 or $-0.465567\dots$ if 2.6941 used) <i>their</i> – $0.4436 > -1.96$ oe M1 dep on award of previous M1	

Question			Answer/Indicative content	Marks	Part marks and guidance		
			insufficient evidence to suggest (at 5% level) that the (population) mean length of time (computers are kept) is not 5 years		use of 0.025, as long as previous M1 awarded may be embedded in conclusion in context	do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; A0 if answer spoiled	
			Total	8			

Question		Answer/Indicative content	Marks	Part marks and guidance		
30	a	$2p + q + 0.2 + 0.3 = 1$ soi oe $2 \times p \times q = 0.06$ soi eliminate p or q with a correct substitution from one of <i>their</i> equations $q^2 - 0.5q + 0.06 = 0$ or $2p^2 - 0.5p + 0.03 = 0$ oe $q = 0.2$ or 0.3 and $p = 0.15$ or 0.1 $(q < 2p \text{ so}) q = 0.2$ and $p = 0.15$	B1 (AO2.1) M1 (AO3.1a) M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) A1 (AO3.2a) [6]	allow M1 if 2 omitted eg $2 \times \frac{0.03}{q} + q = 0.5$ or $2p + \frac{0.03}{p} = 0.5$ may be implied by eg $q = 0.2$ or 0.3 and $2p = 0.3$ or 0.2	NB if 2 omitted, A0 for $2p^2 - 0.5p + 0.6 = 0$ or $2q^2 + q + 0.24 = 0$ which have no real root	
	b	$10 \times q \times (1 - q)^9$ soi 0.27 or 0.268 or awrt 0.2684 isw	M1 (AO1.1) A1 (AO1.1) [2]	FT <i>their</i> q where $0 < q < 1$		

Question		Answer/Indicative content	Marks	Part marks and guidance		
	c	$H_0: p = 0.2$ $H_1: p > 0.2$ p is the probability that the spinner shows a 1 (on any given spin) oe use of $X \sim B(100, 0.2)$ where x is the number of 1s obtained in 100 spins to obtain $P(X \geq k)$ or $P(X \leq k)$ $P(X \leq 27) = \text{awrt } 0.97$ or $P(X \geq 28) = \text{awrt } 0.034$ $0.034 < 0.05$ or $0.97 > 0.95$ significant or reject H_0 or accept H_1; may be embedded in conclusion in context there is sufficient evidence to suggest (at 5% level) that the probability of a score of 1 is greater than 0.2	B1 (AO1.1) B1 (AO2.5) M1 (AO3.3) A1 (AO1.1) M1 (AO3.4) A1 (AO1.1) A1 (AO2.2b) [7]	both hypotheses ; allow equivalent in words or eg $P(1) = 0.2$ $k = 27, 28$ or 29 or critical region is $X \geq 28$ or 28 is in critical region must have the correct probability or correct critical region for the last two A marks	allow any parameter as long as clearly defined as probability M0 for $P(X = k)$ NB $P(X = 28) = 0.014...$ $P(X = 27) = 0.020168...$ FT <i>their</i> probability, dependent on award of first M1 do not allow eg conclude / prove / indicate or other assertive statement instead of suggest; A0 if answer spoiled	
		Total	15			

Question			Answer/Indicative content	Marks	Part marks and guidance		
31	a		Lee is wrong because he should make the comparison of 0.033 with 0.05 he should make the comparison of 0.37154 with 0	B1 (AO2.2a) B1 (AO2.2b) [2]	allow he should have compared r with the critical value	if B0B0 SC1 for Lee has confused r with p or for 0.37154 suggests positive correlation	
	b		$465467 + 2 \times 204356$ awrt 874180 (or 867940 from use of 210236) from scatter diagram the outliers are approximately 920 000, 1 200 000	M1 (AO2.1) A1 (AO2.2b) [2]	condone use of 201236 instead of 204356; ignore work relating to lower tail numerical values must be mentioned	or 521000 + 1.5 × (521000 – 342500) or 788750 in which case accept two or three outliers identified extra one is approximately 800 000	
	c		the pmcc would (probably) be closer to 0 because the scatter is less well modelled by a straight line the p-value would increase because a value which is closer to 0 is more likely assuming there is no correlation	B1 (AO2.2b) B1 (AO2.2b) [2]	if B0B0 allow SC1 for r closer to 0 and p-value larger		
	d		the student's suggestion is reasonable, since there are other regions defined in the LDS	B1 (AO2.2b) [1]			
			Total	7			